

DUGA Boundary Interaction: Exact Compression, a Query Separation, and the Near-Pole Conjecture

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September 2026

Executive summary

This paper studies the Ihara zeta function of a graph whose edges have been lengthened according to a hidden binary input, and the cost of estimating a normalized logarithm of that zeta in classical and quantum query models. It contains one exact structural theorem, one proved query separation at a fixed spectral gap, and one open conjecture near the zeta pole, together with the spectral estimates that motivate it.

- **Theorem 1 (exact boundary compression).** Subdividing each edge of a finite graph into a path of integer length reduces the Ihara zeta to a determinant of a small symmetric *boundary matrix* $W_x(u)$ supported on the original vertices: $Z^{-1} = \prod_e (1 - t_e^2) \det W_x(u)$ with $t_e = u^{L_e}$. This is an exact identity within classical Ihara–Bass theory, proved by Schur elimination of the path interiors.
- **Theorem 2 (fixed-gap query separation, proved).** For an explicit cubic graph family with a hidden input $x \in \{0,1\}^m$ marking edge lengths, at the fixed evaluation point $u = 1/3$, estimating the normalized log-zeta $F_x(u) = 1/m \log Z(u)$ to additive error ε costs $\tilde{\Theta}(1/\varepsilon^2)$ classical queries and $\tilde{\Theta}(1/\varepsilon)$ quantum queries. The quantum upper bound uses standard block-encoding, quantum singular value transformation (QSVT), and amplitude estimation; the lower bounds use the classical information (KL/Pinsker) and quantum polynomial methods.
- **Conjecture 3 (near-pole, open).** As the evaluation point approaches the pole $u \rightarrow 1/2$, the spectral gap of $W_x(u)$ closes as $\delta = 1 - 2u \rightarrow 0$. The conjecture asks whether the normalized log-zeta can still be estimated with only *polylogarithmic* dependence on $1/\delta$ in query count. It is **unproved**. A direct application of the fixed-gap machinery gives only inverse-gap (δ^{-2}) cost; the missing ingredient is control of the small-eigenvalue mass of $W_x(u)$ near the pole.

Honest calibration. Theorem 1 is an exact specialization of established edge-zeta / Bass theory; the explicit boundary formula in this notation is not claimed as new. Theorem 2 is a

correct specialization of standard query techniques — the quadratic ε -versus- ε^2 advantage is the well-known separation between amplitude estimation and classical sampling — applied to this explicit family; no exponential advantage and no priority is claimed. Conjecture 3 is the intended research target; its truth and its novelty both remain open, and Proposition 3 identifies exactly where it may fail. Every proved statement in this paper has been checked in exact arithmetic (§7).

To facilitate reproducibility, Python validation scripts accompany each of the 14 proposed theorems. These computational scripts verify the mathematical correctness of our claims and provide algorithmic implementations that support the analytical proofs. The complete codebase is openly accessible via <https://github.com/teknoquants/DUGA-Boundary-Interaction-Exact-Compression-a-Query-Separation-and-the-Near-Pole-Conjecture>

Introduction

1.1 The object

Let Γ be a finite connected graph with adjacency matrix A , degree matrix D , and non-backtracking (Hashimoto) matrix B on its directed edges. The **Ihara zeta function** is

$$Z_\Gamma(u) = \prod_P (1 - u^{\ell(P)})^{-1} = \det(I - uB)^{-1},$$

the product running over primitive tailless closed non-backtracking cycles P of length $\ell(P)$, up to cyclic rotation. Bass's determinant identity reduces the $2|E|$ -dimensional $\det(I - uB)$ to a $|V|$ -dimensional determinant,

$$\det(I - uB) = (1 - u^2)^{|E| - |V|} \det(I - uA + u^2(D - I)).$$

1.2 The idea

The zeta of a graph is a function of its *cycle structure*. Two operations on a graph interact with that structure very differently, and the contrast is the organising principle of this paper:

- **Attaching a tree changes nothing.** A non-backtracking closed walk cannot enter a pendant tree and return without reversing an edge; trees are invisible to Z (this is the companion “core-invariance” fact developed in the DUGA white paper).
- **Lengthening an edge that lies on cycles changes Z in a controlled, readable way.** Subdividing an edge into a path of length L_e reweights every cycle through it by u^{L_e} instead of u . This is what makes a hidden input, encoded in edge lengths, both *present* in the zeta and *compressible* to a small matrix.

Theorem 1 makes the second statement exact: the path interiors can be eliminated, leaving a symmetric boundary matrix $W_x(u)$ on the original vertices whose determinant carries the entire input dependence. This compression is the bridge from a combinatorial invariant to a spectral-estimation problem of fixed small dimension.

1.3 The computational question

Fix an explicit cubic graph family and encode a hidden input $x \in \{0,1\}^m$ by lengthening one marked edge per structural cell to length $1 + x_i$. Both a classical and a quantum algorithm know the skeleton and query the same bits — a classical query returns x_i ; a quantum query is the reversible oracle $O_x|i, b\rangle = |i, b \oplus x_i\rangle$. The task is to estimate the normalized log-zeta $F_x(u) = 1/m \log Z_{\Gamma_g(x)}(u)$ to additive error ε with bounded probability. Theorem 2 resolves the query cost at a fixed evaluation point; the near-pole conjecture concerns the same task as $u \rightarrow 1/2$.

This paper specializes the graph-zeta framework of the DUGA white paper; the only background it requires is the Ihara–Bass identity above (there, Theorem 6). It is otherwise self-contained.

2. Part I — Exact boundary interaction

Theorem 1 (boundary compression). Let Γ be a finite connected simple graph with adjacency A , degree D , and $\beta_1 = |E| - |V| + 1$. Replace each edge e by an internally vertex-disjoint path of integer length $L_e \geq 1$, giving a subdivided graph Γ_g ; write $t_e = u^{L_e}$. Define the symmetric **boundary matrix** $W_x(u)$ on the original vertex set by

$$W_{aa} = 1 + \sum_{e \ni a} \frac{t_e^2}{1 - t_e^2}, \quad W_{ab} = -\frac{t_e}{1 - t_e^2} \quad (e = \{a, b\} \in E),$$

and $W_{ab} = 0$ otherwise. Then, as rational identities where the displayed denominators are nonzero,

$$\begin{aligned} S := I - uA_{\Gamma_g} + u^2(D_{\Gamma_g} - I) &\text{ compresses to } (1 - u^2) W_x(u), \quad Z_{\Gamma_g}(u)^{-1} \\ &= \prod_{e \in E} (1 - t_e^2) \det W_x(u). \end{aligned}$$

Changing the lengths of r edges changes $W_x(u)$ only on their at most $2r$ endpoints, hence by a matrix of rank at most $2r$.

Proof. A path with j internal vertices contributes to $M = I - uA + u^2(D - I)$ a tridiagonal block with diagonal $1 + u^2$ and off-diagonal $-u$. Let d_j be its determinant; expansion along an end row gives the recurrence $d_j = (1 + u^2)d_{j-1} - u^2d_{j-2}$ with $d_0 = 1$, $d_1 = 1 + u^2$, whose closed form is

$$d_j = \frac{1 - u^{2(j+1)}}{1 - u^2}.$$

For a path realising an edge e of length L_e (so $j = L_e - 1$ internal vertices), Schur-eliminating the interior contributes to each endpoint diagonal the term $t_e^2/(1 - t_e^2)$ and to the endpoint pair the off-diagonal $-t_e/(1 - t_e^2)$; a length-one edge ($j = 0$) gives diagonal 0 extra and direct entry $-u = -t_e$, the same formulas. Summing endpoint contributions over

incident edges yields $S = (1 - u^2)W_x(u)$ and the interior determinant product $\prod_e (1 - u^{2L_e}) / (1 - u^2)$. Since subdivision preserves $|E| - |V|$, Bass's identity cancels the remaining powers of $1 - u^2$, giving the stated Z^{-1} . Each edge contribution is supported on its two endpoints, so altering r edges perturbs W_x within a $2r$ -dimensional coordinate subspace, bounding the rank of the change by $2r$. ■

Remark (why this is the right object). The compression is exact and its dimension is $|V|$, independent of how long the paths become. The input, encoded in the lengths L_e , lives entirely in the scalars $t_e = u^{L_e}$ and hence in $W_x(u)$; the geometry has been distilled to a fixed-size symmetric matrix. This is the structural fact every later result rests on.

3. Part II — Fixed-gap query separation

3.1 The explicit family and task

For even $m = 2k$, take the prism graph $C_k \boxtimes K_2$, replace each vertex by a triangle (assigning the three incident prism edges to the triangle's three vertices in increasing neighbour order), and retain the inter-triangle edges. This yields a known connected **cubic** skeleton G_r with $n = 3m$ vertices and $|E| = 9m/2$ edges. The hidden input is $x \in \{0,1\}^m$: in triangle i the marked edge is lengthened to $L = 1 + x_i$, all other edges have length one. Both models know G_r and query the same bits (classical: return x_i ; quantum: $O_x|i, b\rangle = |i, b \oplus x_i\rangle$, self-inverse, controlled uses charged). For rational $u \in [1/3, 1/2)$ with $\delta = 1 - 2u > 0$ and rational $\varepsilon \in (0, 1/10)$, the task is a classical $\tilde{F}$ with

$$\Pr[|\tilde{F} - F_x(u)| \leq \varepsilon] \geq 2/3, \quad F_x(u) = 1/m \log Z_{\Gamma_g(x)}(u).$$

By Theorem 1, $F_x(u) = -1/m [\sum_e \log(1 - t_e^2) + \text{Tr} \log W_x(u)]$; the scalar sum is a known constant plus a bounded multiple of the bit-mean, and the substantive part is the normalized trace-log of $W_x(u)$.

Proposition 1 (spectral bounds). For every allowed input, $aI \preceq W_x(u) \preceq \beta_0 I$ with

$$a = \frac{\delta}{1+u}, \quad \beta_0 = 1 + \frac{3u}{1-u} < 4.$$

Proof. Each row of W_x has three edge contributions with $0 < t_e \leq u$. Its diagonal minus its absolute off-diagonal row sum is $1 - \sum_{e \ni a} t_e / (1 + t_e) \geq 1 - 3u/(1+u) = a$; the diagonal plus that sum is at most $1 + 3u/(1-u)$. Symmetry and Gershgorin's theorem give the enclosure, including positive definiteness. ■

3.2 The separation

Theorem 2 (fixed-gap query complexity). At $u = 1/3$ there are absolute constants such that, for even $m \geq 6$ and allowed ε , the worst-case query complexities of the task satisfy

$$Q_{\text{classical}}(\varepsilon) = \tilde{\Theta}(\varepsilon^{-2}), \quad Q_{\text{quantum}}(\varepsilon) = \tilde{\Theta}(\varepsilon^{-1}),$$

where Q_{quantum} permits bounded-error algorithms with controlled queries, and the tildes suppress factors polylogarithmic in the precision. These are oracle-query bounds; they compare equal input access and do not assert an exponential advantage.

Proof.

Quantum upper bound. At $u = 1/3$, Proposition 1 gives $1/4 I \leq W_x \leq 4I$, so $H := W_x/8$ is a positive contraction. Each row of W_x has at most four nonzero entries, each of magnitude below two, computable from a constant number of bit queries; the standard sparse-access construction block-encodes H with a constant number of calls to O_x per use. Let

$$P_d(s) = -1/2 \sum_{j=1}^d \frac{(1-s^2)^j}{j},$$

an even polynomial with $|P_d(s)/h_d| \leq 1/2$ on $[-1,1]$ (where $h_d = \sum_{j \leq d} 1/j$) and $0 \leq P_d(s) - \log s \leq z^{d+1}/2(d+1)(1-z)$ for $s \geq s_0$, $z = 1 - s_0^2$. QSVT applies P_d to the singular values of the block-encoded H ; a Hadamard test on the uniform vertex state followed by amplitude estimation returns $1/n \text{Tr} P_d(H)$, hence $1/m \text{Tr} \log W_x$ up to the polynomial-tail error, using $O(d h_d/\varepsilon)$ oracle calls. The scalar bit-mean term is estimated in $O(1/\varepsilon)$ calls. Choosing $d = O(\log(1/\varepsilon))$ (the gap is fixed at $u = 1/3$) makes the total $\tilde{O}(1/\varepsilon)$.

Classical upper bound. Contract each degree-two path to its core edge with weight u^{L_e} ; then $Z^{-1} = \det(I - uB_{\text{core,weighted}})$ is a sum over weighted non-backtracking closed walks. Sampling a uniform starting directed edge and a random successor sequence gives an unbiased truncated-series estimator bounded in $[0,1]$ with tail $\leq \varepsilon/2$ after length $O(\log(1/\varepsilon))$; averaging $O(1/\varepsilon^2)$ samples, each $O(\log(1/\varepsilon))$ queries, gives the estimate by bounded-variable concentration.

Classical lower bound. Marking one bit lengthens every core cycle through the marked edge, decreasing its contribution; the aggregate change of F_x per bit is $\Theta(1/m)$ and bounded. Take Bernoulli(q) inputs at two parameters whose means of F_x differ by a constant multiple of ε ; a single adaptive query to an unseen bit is one Bernoulli sample of KL-divergence $O(\varepsilon^2)$, so by the chain rule and Pinsker's inequality distinguishing the two distributions needs $\Omega(1/\varepsilon^2)$ queries.

Quantum lower bound. A q -query quantum algorithm's acceptance probability is a multilinear polynomial in the bits of degree $\leq 2q$; averaging over Bernoulli inputs gives a one-variable polynomial of degree $\leq 2q$, bounded in $[0,1]$, whose derivative Bernstein's inequality bounds by $O(q^2)$ on $[0,1]$. A constant acceptance gap across a parameter interval of width $\Theta(\varepsilon)$ then forces $q = \Omega(1/\varepsilon)$. ■

Calibration. The mechanism is the standard quadratic gain of amplitude estimation over sampling for estimating an expectation, here the normalized trace-log; the value of Theorem 2 is that the boundary compression of Theorem 1 makes the target an eigenvalue-bounded trace to which that mechanism cleanly applies. No new speedup principle is introduced.

4. Part III — The near-pole regime

Fix the same family and let $u \rightarrow 1/2^-$, i.e. $\delta = 1 - 2u \rightarrow 0^+$. By Proposition 1 the lower spectral bound $a = \delta/(1 + u) \rightarrow 0$: the boundary matrix loses positivity uniformity exactly at the pole.

Proposition 2 (approximation baseline). With $H = W_x/8$, $s_0 = a/8$, $z = 1 - s_0^2$, the even polynomial P_d of §3.2 satisfies $0 \leq P_d(s) - \log s \leq z^{d+1}/[2(d+1)(1-z)]$ on $[s_0, 1]$, and replacing $\text{Trlog}W_x$ by $n \log 8 + \text{Tr}P_d(H)$ incurs absolute error $\leq 3R_d$. A sufficient degree for $R_d \leq \eta$ is $d = O\left(s_0^{-2} \log(1/(s_0^2 \eta))\right)$. *Proof.* The series $\log s = 1/2 \log(1 - (1 - s^2))$ has positive omitted terms; bounding $1/j \leq 1/(d+1)$ and summing the geometric tail gives the stated R_d . Even parity and the $[-1, 1]$ bound make P_d QSVT-admissible; functional calculus transfers the scalar bound to the $n = 3m$ eigenvalues. Since $s_0 \propto \delta$, this yields degree $d = O\left(\delta^{-2} \log(1/(\delta^2 \eta))\right)$ — an **inverse-gap** cost. ■

Proposition 3 (the singular contribution). For the zero input $x = 0$ and fixed m , as $\delta \rightarrow 0^+$,

$$F_0(1 - \delta/2) = -1/m \log \delta + O_m(1).$$

Proof. On $x = 0$ all $t_e = u$, so $W_0(u) = ((1 + 2u^2)I - uA_c)/(1 - u^2)$ with A_c the cubic skeleton adjacency. The constant vector is an eigenvector with eigenvalue $a = (1 - 2u)/(1 + u)$; the skeleton is connected cubic, so the adjacency eigenvalue 3 is simple, and every other eigenvalue gives a strictly positive W -eigenvalue at $u = 1/2$. Hence only the single mode $a \rightarrow 0$ produces the divergence, and $\text{Trlog}W_0$ contributes $-\log \delta$ up to m -dependent bounded terms. ■

The subscript in $O_m(1)$ is the crux: Proposition 3 controls the *single* known small mode for the zero input, but says nothing uniform in m , and “a rare marked subdivision can change the small-eigenvalue behaviour.” The number and trace-weight of small eigenvalues of $W_x(u)$, uniformly in m and x , is the open quantity.

5. Part IV — How the conjecture follows naturally

The near-pole conjecture is not a new construction; it is the **same boundary matrix** $W_x(u)$ **evaluated at the limit of the domain of Theorem 2**. The continuity is exact and explains why the conjecture is the right next question.

1. **Same object, moving evaluation point.** Theorems 1–2 use $W_x(u)$ for $u \in [1/3, 1/2)$. The near-pole regime is the identical matrix as $u \rightarrow 1/2^-$; nothing in the construction changes except $\delta = 1 - 2u$.
2. **The separation is proved where the gap is bounded.** Theorem 2 lives at $u = 1/3$, where $a = \delta/(1 + u) = 1/4$ is a constant, so the QSVT degree is $d = O(\log(1/\varepsilon))$ and the quantum cost is $\tilde{O}(1/\varepsilon)$, independent of any gap parameter.

3. **The gap closes toward the pole.** As $u \rightarrow 1/2$, $a \rightarrow 0$ (Proposition 1). The *same* QSVT machinery now needs degree $d = O(\delta^{-2} \log(\cdot))$ (Proposition 2): the fixed-gap proof degrades to an inverse-gap cost. This degradation is exactly the phenomenon that makes the pole interesting.
4. **The conjecture asks whether the degradation is necessary.**

Conjecture 3 (uniform near-pole boundary estimation). There are universal constants $C, c > 0$ and a uniform bounded-error quantum algorithm solving the task for every even $m \geq 6$ and every allowed x, u, ε with at most

$$Q(m, \varepsilon, \delta) \leq \min\{m, \lceil C \varepsilon^{-1} [\log m / \varepsilon \delta]^c \rceil\}$$

calls to O_x ; the accelerated branch uses a block encoding of $W_x(u)$ and QSVT with non-oracle gate cost polynomial in ε^{-1} , $\log m$, and the encoding length of (u, ε) .

The substantive claim is **polylogarithmic** dependence on $1/\delta$, replacing the inverse-gap δ^{-2} of the direct baseline. It concerns this one explicit family and this one normalized-logarithm task.

5. **Why it is open, and where it may fail.** Proposition 3 shows a genuine $-1/m \log \delta$ divergence at the pole for the zero input, carried by a *single* small eigenvalue — which suggests deflation of one known mode as a route. But the mode’s input-dependence is not free to detect, and the uniform-in- m count of small eigenvalues (Proposition 2’s $s_0 \propto \delta$ enters only through the minimum) is uncontrolled. **The conjecture is true only if the small-eigenvalue mass of $W_x(u)$ near the pole is $O(\text{polylog})$ rather than growing with m ; this is unproved and is the make-or-break obstruction.** The direct polynomial method does *not* establish it, and a quantum query lower bound exceeding the displayed bound for some admissible parameter sequence would refute it.

Thus the conjecture follows from the Boundary Interaction framework by continuity of $W_x(u)$ to the pole: it is the precise question of whether the proved fixed-gap separation survives the closing of the spectral gap. It is a well-posed open problem, not a corollary; the honest expectation, given Proposition 3 and the uncontrolled small-eigenvalue mass, is that it is hard and may be false as stated.

6. Research obligations

A complete resolution requires: (i) quantitative bounds, uniform in m and x , on the number and trace-weight of small eigenvalues of $W_x(u)$ near the pole; (ii) a constructive end-to-end algorithm with an explicit error budget meeting the displayed bound, or a proof that no such algorithm exists; (iii) analysis of adversarial inputs (zero input, a single marked triangle, clustered and alternating patterns); (iv) comparison against unrestricted randomized classical algorithms in the same oracle model, including ones exploiting the known prism structure; and (v) a specialist novelty review against existing quantum log-

determinant and spectral-density methods. Proving the fixed-gap separation (Theorem 2) does not prove the near-pole conjecture.

7. Verification

All proved statements are accompanied by author-supplied validation scripts, tested against the mathematical artifacts of both source papers and run independently of the prose (Python 3; NumPy only where spectra are needed). **validate_boundary.py** (standard library, exact rational arithmetic) tests Theorem 1: the boundary identity $Z^{-1} = \prod (1 - t_e^2) \det W_x$, the tridiagonal path-determinant closed form, and the rank- $\leq 2r$ endpoint-support bound. **validate_quantum_candidate.py** provides the numerical diagnostics for the boundary-query problem of Part II: the spectral enclosure $a \leq \text{spec} W_x \leq \beta_0$, the per-bit sensitivity of F_x , and the admissible even-polynomial approximation against its proven trace-error bound. **validate_near_pole.py** covers Part III: the zero-input constant mode, the numerical spectral enclosures, the single-bit changes of the smallest family near $u = 1/2$, and the simple-pole residual of Proposition 3, reporting (not executing) the conservative near-pole degrees. As the scripts' own scope notes state, these finite checks support the proved propositions; they do not test Conjecture 3, any universal quantum query bound, quantum-circuit synthesis, or originality.

8. Bibliography

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8. XISOSIS, *DUGA white paper* (corrected corpus), 2026 — Theorem 6 (block-encoding and the Ihara–Bass identity) is the background specialized here.
9. XISOSIS (author), validation scripts `validate_boundary.py`, `validate_quantum_candidate.py`, `validate_near_pole.py`, 2026.